

Statistics of Financial Markets

Chapter 7: Efficient markets, random walk and variance-ratio tests



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AMH**: Adaptive Markets Hypothesis
- **AR**: AutoRegressive (model)
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CAPM**: Capital Asset Pricing Model
- **CD**: Chow–Denning (multiple variance-ratio test)
- **CME**: Chicago Mercantile Exchange
- **COVID-19**: COronaVIrus Disease 2019
- **EMH**: Efficient Market Hypothesis
- **EODHD**: EOD Historical Data (market-data provider)
- **EU**: European Union
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)

Acronyms used in this material (2/2)

- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GMM**: Generalized Method of Moments
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (stationarity test)
- **LB**: Ljung–Box (test)
- **MA**: Moving Average (model)
- **MIT**: Massachusetts Institute of Technology
- **OLS**: Ordinary Least Squares
- **PP**: Phillips–Perron (unit-root test)
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **RW1**: Random Walk 1 (i.i.d. increments)
- **RW2**: Random Walk 2 (independent increments)
- **RW3**: Random Walk 3 (uncorrelated increments)
- **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- **VR**: Variance Ratio
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's Question and Route

- **Question**: can yesterday's prices tell us anything about tomorrow's return?
  - if yes, a simple rule could beat the market; if no, the market is efficient in the weak sense
  - Chapter 4 defined the random walk and the martingale; here we test them on real prices
- **Route** of the chapter
  - efficient markets: three forms, the joint-hypothesis problem
  - random walk hypotheses RW1, RW2, RW3 and the martingale
  - white noise and the autocorrelation function; Ljung–Box and runs tests
  - unit roots: ADF, Phillips–Perron, KPSS; prices against returns
  - variance-ratio tests; efficiency across markets and over time; anomalies

Learning Outcomes

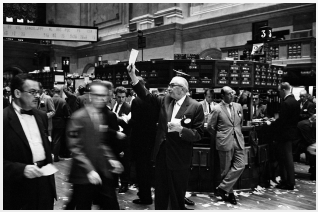
- State the three forms of market efficiency and explain why every test is a joint test
- Tell RW1, RW2, RW3 and the martingale apart, and say which of them volatility clustering violates
- Read a sample ACF with the right confidence band and compute Box–Pierce, Ljung–Box and runs statistics
- Run and combine ADF, Phillips–Perron and KPSS tests on prices and on returns
- Compute a variance ratio and its robust test, for one horizon and for several horizons together
- Judge, on real data, which markets look efficient, and whether this changes over time

Reading and Tools

- Textbook: Franke, Härdle & Hafner (2019), Ch. 11 (11.3 expectations and efficient markets, 11.5 random walk tests, 11.6 unit root tests)
  - exercises with solutions: Borak, Härdle & López-Cabrera (2013), Ch. 11
- Econometrics of predictability: Campbell, Lo & MacKinlay (1997), Ch. 2 (random walks, variance ratios) and Ch. 4 (event studies)
- Python Quantlets of this chapter: Quantlets/Ch_07
  - ported from the SFE Quantlets SFEtimewn, SFEacfar1, SFEacfar2, SFEacfma1 and SFEacfma2
- Lecture notebook: open in Google Colab
- Video course: Efficiency of cryptocurrency markets – A GMM-based analysis

Can You Beat the Market?

- If past prices predicted future returns, a simple rule would earn money with no risk
  - traders would use the rule; their trades would move prices until the pattern disappears
- The evidence on professional investors
  - mutual funds, on average, did not beat a buy-and-hold portfolio after costs Jensen (1968)
  - few funds show skill large enough to cover their costs Fama & French (2010)
- A survey of the arguments for and against: Malkiel (2003)



The trading floor of the New York Stock Exchange, 1963

Photo: Thomas J. O'Halloran (1963), Library of Congress; public domain; Wikimedia Commons

1900: Prices as a Random Walk

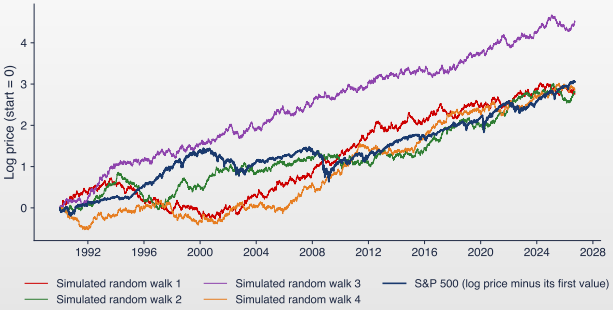
- Louis Bachelier, *Théorie de la spéculation* Bachelier (1900)
  - the first mathematical model of prices: the change of a price is a sum of many small random shocks
  - the expected gain of the speculator is zero
- Kendall (1953): 22 British price series behave like a "wandering" series
  - the changes from one week to the next look like independent draws
- Roberts (1959): simulated random walks show "patterns" that chartists see in real prices



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons

Which One Is the S&P 500?



- Four random walks with i.i.d. Normal steps (mean 0.033%, standard deviation 1.13% per day, as the S&P 500) and the S&P 500 log price since 1990
- **Question for the room**: without the legend, could you tell which path is real?
- **Answer**: hardly; booms and long falls appear in pure noise too (end values: 2.76 to 4.50; S&P 500: 3.06)

SFM_ch7_random_walk

1965: Properly Anticipated Prices Fluctuate Randomly

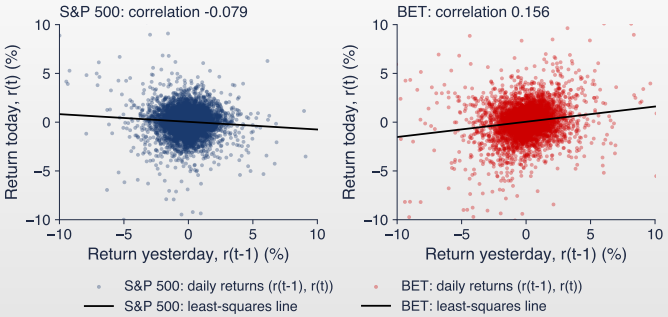
- Paul Samuelson Samuelson (1965): if prices already contain all that traders expect, the next change is unpredictable
  - formally: the price is a **martingale**, $E[P_{t+1} \mid \mathcal{F}_t] = P_t$ (Chapter 4)
  - $\mathcal{F}_t$: the information available at time t
- Randomness of price changes is a sign of a market that works, not of chaos
- Fama (1965): daily returns of the 30 Dow Jones stocks show almost no autocorrelation



Paul A. Samuelson (1915–2009), Nobel Prize 1970

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

Does Yesterday Predict Today?



- Each point: (return yesterday, return today); the line: least squares; its slope is the first autocorrelation $\hat{\rho}(1)$
- S&P 500: $\hat{\rho}(1) = -0.079$ (a slight reversal); BET: $\hat{\rho}(1) = 0.156$ (a slight continuation); are these numbers different from zero?

SFM_ch7_random_walk

What Is an Efficient Market?

- **EMH** (efficient market hypothesis) Fama (1970): prices "fully reflect" the available information Ω_t
  - no trading rule based on Ω_t earns more than the return that compensates its risk
  - prices move only on **news**: information that was not in Ω_t
- Efficiency is relative to an information set
  - the smaller Ω_t, the weaker the claim and the easier the test
- Efficient does not mean "returns are zero": the expected return pays for risk



Eugene F. Fama, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

Three Forms of Efficiency

Form	Information set Ω_t	What cannot earn excess returns	Typical test
Weak	past prices and returns	technical analysis, chart patterns	autocorrelation, runs, variance ratio
Semi-strong	all public information	fundamental analysis of public data	event studies
Strong	all information, also private	insider information	returns of insiders and fund managers

- The forms are nested: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak; a rejection of the weak form rejects all three
- This chapter tests mainly the weak form: past prices are free data for everyone Fama (1970), Fama (1991)

The Fair-Game Model

- Let r_{t+1} be the return of an asset and $E[r_{t+1} \mid \Omega_t]$ its **equilibrium expected return** (the return that pays for its risk)
  - the **excess return** (surprise): $z_{t+1} = r_{t+1} - E[r_{t+1} \mid \Omega_t]$
- **Fair game**: $E[z_{t+1} \mid \Omega_t] = 0$
  - forecast errors are zero on average whatever we know at time t
  - a rule that invests $w(\Omega_t)$ earns $E[w(\Omega_t)\, z_{t+1}] = 0$ in excess of equilibrium
- With a constant expected return μ: $E[r_{t+1} \mid \Omega_t] = \mu$; returns minus μ are a **martingale difference**
- Nothing here says the variance is constant or the distribution is Normal

The Joint-Hypothesis Problem

- To test efficiency we must know the equilibrium expected return $E[r_{t+1} \mid \Omega_t]$
  - this needs a model of equilibrium: constant mean, CAPM (capital asset pricing model), factor models
- Every test of efficiency is a **joint test** of efficiency and of the model Fama (1991)
  - a rejection may mean an inefficient market, or a wrong model of expected returns
  - efficiency alone can never be rejected
- For daily returns the problem is small: daily expected returns are tiny (about 0.033% for the S&P 500), so any reasonable model gives almost the same answer

The Grossman–Stiglitz Paradox

- Information is costly to collect and to analyse Grossman & Stiglitz (1980)
  - if prices reflected all information perfectly, nobody would be paid for collecting it
  - then nobody would collect it, and prices could not reflect it
- Conclusion: a market can only be **nearly** efficient
  - small inefficiencies pay informed traders for their costs
  - the question becomes "how efficient, where and when?", not "yes or no?"
- This view leads to the adaptive markets hypothesis of Section 8

2013: One Nobel Prize, Two Views



Eugene F. Fama: prices are hard to predict over days and weeks

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Robert J. Shiller: prices swing too much compared with fundamentals

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Both are right, at different horizons: short-run returns are almost unpredictable; long-run returns are partly predictable from valuation ratios

The Random Walk Model of Log Prices

- Log price $p_t = \ln P_t$; **random walk with drift** μ: $p_t = \mu + p_{t-1} + \varepsilon_t$
  - the return $r_t = p_t - p_{t-1} = \mu + \varepsilon_t$: the increments ε_t have mean 0
  - by recursion: $p_t = p_0 + \mu t + \sum_{s=1}^{t} \varepsilon_s$: every shock is **permanent**
- If the ε_t are uncorrelated with variance σ^2:
  - $E[p_t] = p_0 + \mu t$ and $\mathrm{Var}(p_t - p_0) = t\sigma^2$: the uncertainty grows with the horizon
  - the q-day return $r_t(q) = r_t + \cdots + r_{t-q+1}$ has $\mathrm{Var}(r_t(q)) = q\sigma^2$
- Log prices, not prices: P_t cannot become negative, and log returns add over time (Chapter 1)

Three Random Walk Hypotheses

- **RW1** (i.i.d. increments): ε_t independent and identically distributed, $\varepsilon_t \sim \mathrm{IID}(0, \sigma^2)$ Campbell, Lo & MacKinlay (1997)
  - the strongest: no function of the past predicts any function of the future
- **RW2** (independent increments): ε_t independent, but not identically distributed
  - allows the variance to change over time in a way that is not predictable from the past (regimes)
- **RW3** (uncorrelated increments): $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ for all $k \neq 0$
  - the weakest: allows $\mathrm{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$, that is, volatility clustering
- Nested: RW1 $\Rightarrow$ RW2 $\Rightarrow$ RW3; tests of RW3 use only autocorrelations of returns

What Each Hypothesis Allows

Property of the returns	RW1	RW2	RW3	**Martingale**
$\mathrm{Corr}(r_t, r_{t-k}) = 0$	yes	yes	yes	yes
$E[r_t \mid \text{past}] = \mu$ (unpredictable mean)	yes	yes	not always	yes
constant variance	yes	no	no	no
volatility clustering allowed	no	no	yes	yes
GARCH returns (Chapter 9) satisfy it	no	no	yes	yes

- **GARCH** (generalised autoregressive conditional heteroskedasticity): the variance of tomorrow depends on today's shocks; the mean stays unpredictable
- Weak-form efficiency needs only a martingale (or RW3), not RW1

Martingale and Random Walk

- $\{p_t - \mu t\}$ is a **martingale** if $E[p_{t+1} - \mu(t+1) \mid \mathcal{F}_t] = p_t - \mu t$ Samuelson (1965)
  - equivalently, $E[r_{t+1} - \mu \mid \mathcal{F}_t] = 0$: the returns minus μ are a martingale difference
  - a martingale difference with finite variance is uncorrelated: martingale $\Rightarrow$ RW3
- RW1 and RW2 imply the martingale; the converse fails: a martingale may have predictable variance
- **Question for the room**: does volatility clustering contradict weak-form efficiency?
  - **Answer**: no; it rejects RW1 and RW2, but a predictable variance gives no predictable return; the martingale and RW3 survive

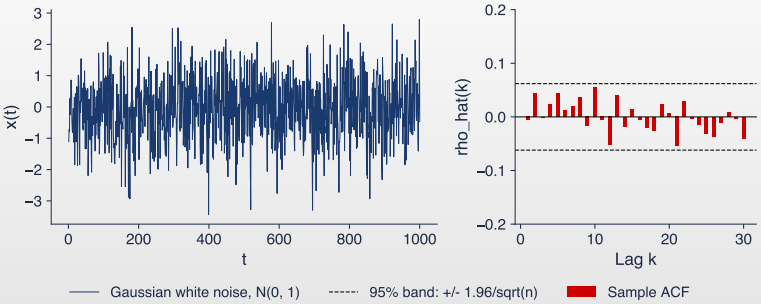
Worked Example: How Risk Grows with the Horizon

- S&P 500 since 1990: daily standard deviation $\hat{\sigma} = 1.1337\%$
  - under a random walk, $\mathrm{sd}(r_t(q)) = \sigma\sqrt{q}$
- Step by step
  - one week, $q = 5$: $1.1337 \times \sqrt{5} = 1.1337 \times 2.236 = 2.53\%$
  - one month, $q = 20$: $1.1337 \times 4.472 = 5.07\%$
  - one year, $q = 250$: $1.1337 \times 15.811 = 17.92\%$
- This is the "square-root-of-time rule" used to scale VaR and volatility
- If returns are autocorrelated, the rule fails: the variance-ratio test of Section 7 measures by how much

White Noise and Autocorrelation

- **Autocovariance** at lag k: $\gamma(k) = \mathrm{Cov}(x_t, x_{t-k})$; **autocorrelation**: $\rho(k) = \gamma(k)/\gamma(0)$
  - the **ACF** (autocorrelation function): $k \mapsto \rho(k)$, $k = 1, 2, \ldots$
- **White noise**: mean 0, constant variance σ^2, $\rho(k) = 0$ for every $k \neq 0$
  - i.i.d. noise is white noise; Gaussian white noise is i.i.d. $N(0, \sigma^2)$
  - white noise need not be independent: GARCH returns are white noise but not i.i.d.
- **Sample ACF**: $\hat{\rho}(k) = \sum_{t=k+1}^{T}(x_t - \bar{x})(x_{t-k} - \bar{x}) \big/ \sum_{t=1}^{T}(x_t - \bar{x})^2$
  - for i.i.d. data, $\hat{\rho}(k) \approx N(0, 1/T)$: 95% band $\pm 1.96/\sqrt{T}$

Gaussian White Noise and Its Sample ACF



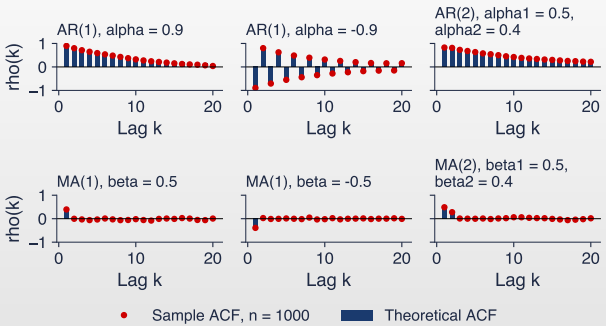
- Left: 1 000 draws from $N(0,1)$ (as in SFEtimewn); right: $\hat{\rho}(1), \ldots, \hat{\rho}(30)$ with the band $\pm 1.96/\sqrt{1000} = \pm 0.062$
- Here 0 of 30 autocorrelations fall outside the band; with 30 lags we expect about 1.5 by chance alone

SFM_ch7_white_noise_acf

The ACF of AR and MA Processes

- **AR(1)** (autoregressive): $x_t = \alpha x_{t-1} + \varepsilon_t$, $|\alpha| < 1$: $\rho(k) = \alpha^k$
  - geometric decay; alternating signs if $\alpha < 0$; example: $\alpha = 0.9 \Rightarrow \rho(3) = 0.9^3 = 0.729$
- **AR(2)**: $x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \varepsilon_t$: $\rho(1) = \alpha_1/(1-\alpha_2)$, $\rho(k) = \alpha_1\rho(k-1) + \alpha_2\rho(k-2)$
  - $\alpha_1 = 0.5$, $\alpha_2 = 0.4$: $\rho(1) = 0.5/0.6 = 0.833$, $\rho(2) = 0.5 \times 0.833 + 0.4 = 0.817$
- **MA(1)** (moving average): $x_t = \varepsilon_t + \beta\varepsilon_{t-1}$: $\rho(1) = \beta/(1+\beta^2)$, $\rho(k) = 0$ for $k \geq 2$
  - $\beta = 0.5$: $\rho(1) = 0.5/1.25 = 0.40$; an MA($q$) has an ACF that cuts off after lag q
- The shape of the ACF tells the type of dependence: slow decay (AR), cut-off (MA), nothing (white noise)

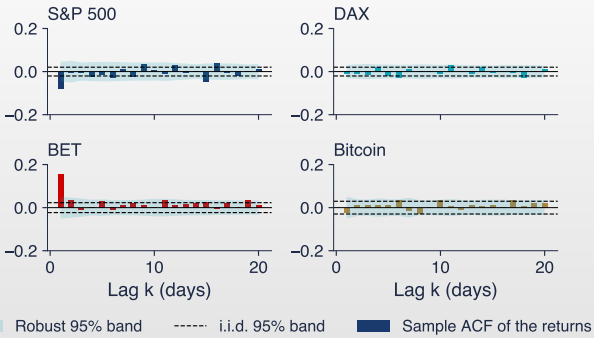
Theoretical and Sample ACF of Six Processes



- Bars: theoretical $\rho(k)$; points: sample ACF of one simulated path of 1 000 values (SFEacfar1, SFEacfar2, SFEacfma1, SFEacfma2)
- A random walk is the limit $\alpha \to 1$ of the AR(1): its ACF would not decay at all

SFM_ch7_white_noise_acf

The ACF of Daily Returns



- Bars: $\hat{\rho}(1), \ldots, \hat{\rho}(20)$; dashed: the i.i.d. band $\pm 1.96/\sqrt{T}$; shaded: the robust band (next slide)

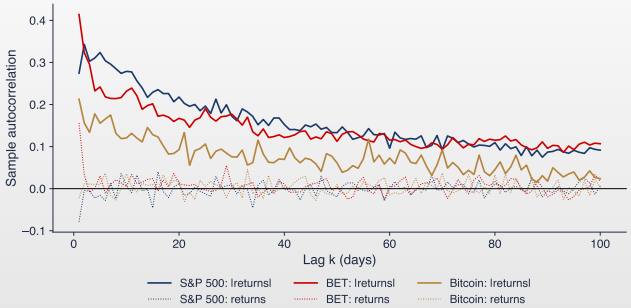
SFM_ch7_autocorrelation_tests

Reading the ACF of Returns

- First autocorrelations: S&P 500 -0.079, DAX -0.009, BET 0.156, Bitcoin -0.022
  - only the S&P 500 and the BET stand out at lag 1, with opposite signs
- The **robust** band: $\pm 1.96\sqrt{\hat{\delta}(k)/T}$, $\hat{\delta}(k) = T\sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$, $e_t = r_t - \bar{r}$ Lo & MacKinlay (1988)
  - i.i.d. returns: $\hat{\delta}(k) \approx 1$; volatility clustering: large returns follow large returns, so $\hat{\delta}(k) > 1$
  - S&P 500 at lag 1: ± 0.046 instead of ± 0.020
- S&P 500, lags outside the band out of 20: 9 with the i.i.d. band, 3 with the robust band
- The i.i.d. band tests RW1; the robust band tests RW3, the hypothesis that matters for efficiency

SFM_ch7_autocorrelation_tests

Returns Are Almost Uncorrelated, Their Size Is Not



- ACF of $|r_t|$: S&P 500 0.27 at lag 1, 0.22 at lag 20, 0.09 at lag 100; the returns themselves: dotted lines near 0
- Large moves follow large moves: RW1 and RW2 are rejected by eye; RW3 and the martingale are not touched by this chart

SFM_ch7_autocorrelation_tests

Portmanteau Tests: Box–Pierce and Ljung–Box

- H_0: $\rho(1) = \cdots = \rho(m) = 0$; H_1: at least one $\rho(k) \neq 0$; one test for m lags together
  - "portmanteau": a suitcase that carries many autocorrelations at once
- Box–Pierce Box & Pierce (1970): $Q_{BP}(m) = T \sum_{k=1}^{m} \hat{\rho}(k)^2$
  - T: the number of returns; m: the number of lags tested; squares, so positive and negative autocorrelations both count
- **LB** (Ljung–Box) Ljung & Box (1978): $Q_{LB}(m) = T(T+2) \sum_{k=1}^{m} \frac{\hat{\rho}(k)^2}{T-k}$, closer to χ^2 in small samples
  - under i.i.d. returns both $\sim \chi^2(m)$; reject H_0 if $Q > \chi^2_{0.95}(m)$; $\chi^2_{0.95}(10) = 18.31$
- Choice of m: here $m = 10$ (two trading weeks); a larger m dilutes a short-lived effect

Worked Example: Ljung–Box Step by Step

- $T = 1000$ daily returns, $\hat{\rho}(1) = 0.06$, $\hat{\rho}(2) = -0.04$, $\hat{\rho}(3) = 0.03$; test with $m = 3$
  - $T\hat{\rho}(k)^2$: 3.6, 1.6, 0.9; their sum: $Q_{BP}(3) = 6.10$
  - $Q_{LB}(3) = 1000 \times 1002 \times (0.06^2/999 + 0.04^2/998 + 0.03^2/997) = 6.12$
- Decision: $\chi^2_{0.95}(3) = 7.81 > 6.12$: do not reject H_0 at 5%
  - note: $\hat{\rho}(1) = 0.06$ lies just inside the band $\pm 1.96/\sqrt{1000} = \pm 0.062$; the joint test does not reject either
- The same numbers with $T = 5000$: Q five times larger, a clear rejection: small autocorrelations become significant in long samples

A Portmanteau Test Robust to Volatility Clustering

- The χ^2 law of Q_{LB} needs $\mathrm{Var}(\hat{\rho}(k)) = 1/T$: true for i.i.d. returns, false with volatility clustering
  - then Q_{LB} rejects a true martingale far too often
- **Robust** statistic: divide each $\hat{\rho}(k)^2$ by its own variance $\hat{\delta}(k)/T$ Escanciano & Lobato (2009)
  - $\tilde{Q}(m) = \sum_{k=1}^{m} \frac{T\hat{\rho}(k)^2}{\hat{\delta}(k)} \sim \chi^2(m)$ for a martingale difference with finite fourth moment
  - with i.i.d. returns $\hat{\delta}(k) \approx 1$ and $\tilde{Q} \approx Q_{BP}$
- Rule: test RW1 with Q_{LB}; test the martingale (weak-form efficiency) with $\tilde{Q}$

Autocorrelation Tests on Real Data

	T	$\hat{\rho}(1)$	$Q_{LB}(10)$	p	$\tilde{Q}(10)$	p	$Q_{LB}(10)$, r^2	runs z	p
S&P 500	9 240	−0.079	90.8	<0.001	18.8	0.043	8015	2.97	0.003
DAX	9 288	−0.009	21.8	0.016	8.6	0.571	3747	1.73	0.084
BET	7 243	0.156	198.3	<0.001	42.9	<0.001	2500	−7.75	<0.001
Bitcoin	4 384	−0.022	20.2	0.028	11.9	0.294	213	2.68	0.007
TLV	3 788	0.010	29.1	0.001	13.2	0.210	481	−1.68	0.093
SNP	3 815	−0.015	20.7	0.023	9.9	0.446	505	1.72	0.085
BRD	3 771	0.032	19.9	0.030	10.8	0.375	416	−3.04	0.002
TGN	3 725	0.055	28.7	0.001	12.3	0.263	639	1.49	0.136

- Classic LB: all eight series reject at 5%; robust $\tilde{Q}$: only the S&P 500 (p-value: 0.043) and the BET (p-value: <0.001)
- Squared returns: huge Q_{LB} everywhere: RW1 and RW2 are rejected for every series (volatility clustering)
- Start of the series: S&P 500 and DAX 1990; BET 1997; Bitcoin 2014; BVB stocks 2010 (TLV without 30–31 May 2016, a data error, Chapter 2)

SFM_ch7_autocorrelation_tests

The Runs Test

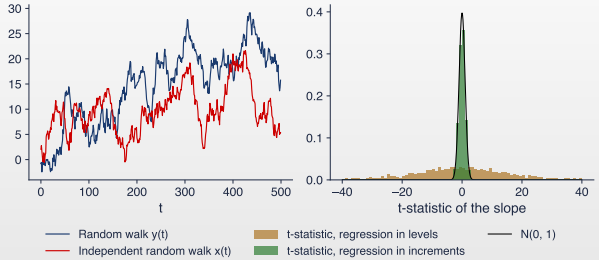
- A **run**: a sequence of returns with the same sign; $++-+++--$ has 4 runs Wald & Wolfowitz (1940)
  - R: the observed number of runs
  - the test uses only the signs: it is **nonparametric**, insensitive to heavy tails and to volatility
- With n_1 positive and n_2 negative returns, $n = n_1 + n_2$, under independence:
  - $E[R] = \dfrac{2n_1n_2}{n} + 1, \quad \mathrm{Var}(R) = \dfrac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}, \quad z = \dfrac{R - E[R]}{\sqrt{\mathrm{Var}(R)}} \approx N(0,1)$
- Too few runs ($z < 0$): signs persist, positive dependence; too many runs ($z > 0$): signs alternate
  - BET: 3 278 runs against 3 606 expected, $z = -7.75$; S&P 500: 4 738 against 4 596, $z = 2.97$
  - the same directions as $\hat{\rho}(1)$: continuation in the BET, reversal in the S&P 500

SFM_ch7_autocorrelation_tests

Stationary and Integrated Series

- **(Weakly) stationary** series: constant mean, constant variance, $\mathrm{Cov}(x_t, x_{t-k})$ depends only on k
  - shocks fade away; the series returns to its mean; notation $I(0)$
- **Integrated of order 1**, $I(1)$: not stationary, but its first difference $\Delta x_t = x_t - x_{t-1}$ is
  - the random walk is $I(1)$: p_t wanders, $\Delta p_t = r_t$ is stationary
- **Unit root**: in $p_t = \phi\, p_{t-1} + \varepsilon_t$ the case $\phi = 1$; $|\phi| < 1$ gives a stationary AR(1)
  - the name: the root of $1 - \phi z = 0$ is $z = 1/\phi = 1$
- Question of this section: are log prices $I(1)$ and returns $I(0)$?

Why It Matters: Spurious Regression



- Regress one random walk on another, independent one ($T = 500$, 2 000 repetitions) Granger & Newbold (1974)
  - $|t| > 1.96$ in 90% of cases, median $R^2 = 0.17$: a "relation" that does not exist
- On the increments: $|t| > 1.96$ in 5.5% of cases, as it should be
  - lesson: test for unit roots before regressing levels ($|t| > 40$ in 2.6% of cases, not drawn)

SFM_ch7_unit_roots

The Dickey–Fuller Test

- Subtract p_{t-1}: $\Delta p_t = c + \gamma\, p_{t-1} + \varepsilon_t$ with $\gamma = \phi - 1$ Dickey & Fuller (1979)
  - $\Delta p_t = p_t - p_{t-1}$; c: a constant; $\gamma < 0$ pulls p_t back towards a level
  - H_0: $\gamma = 0$ (unit root); H_1: $\gamma < 0$ (stationary); a one-sided test
  - statistic: $\tau = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$, the usual t-ratio of OLS (ordinary least squares)
- Under H_0, τ does **not** follow a t or Normal law: its critical values are more negative
  - 5% critical values MacKinnon (1996): -2.86 with a constant, -3.41 with a constant and a trend (Normal: -1.645)
- Three specifications: no constant; constant c; constant and trend $c + bt$
  - log prices drift upwards: use constant and trend; returns: constant only

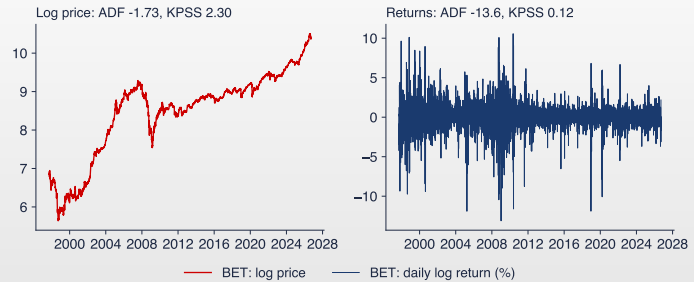
The Augmented Test and a Worked Example

- **ADF** (augmented Dickey–Fuller) Said & Dickey (1984): add lagged differences to remove autocorrelation from ε_t
  - $\Delta p_t = c + bt + \gamma\, p_{t-1} + \sum_{j=1}^{L} \delta_j \Delta p_{t-j} + \varepsilon_t$; same τ, same critical values
  - bt: a linear trend; δ_j: the coefficients of the L lagged differences
  - L chosen by AIC (Akaike information criterion, Chapter 6), at most $12(T/100)^{1/4}$
- **Worked example**: BET log price, constant, no lags, $T = 7\,243$
  - OLS: $\hat{\gamma} = -0.0000138$, $\mathrm{SE}(\hat{\gamma}) = 0.000158$
  - $\tau = -0.0000138/0.000158 = -0.09 > -2.86$: do not reject the unit root
  - here even a usual t-test (critical value -1.645) would not reject; but a τ between -2.86 and -1.645 would be rejected by the wrong (Normal) critical value and kept by the right one

Phillips–Perron and KPSS

- **PP** (Phillips–Perron) Phillips & Perron (1988): the Dickey–Fuller regression without lags; τ is corrected for autocorrelation and heteroskedasticity
  - the correction uses a long-run variance with Bartlett weights Newey & West (1987); same H_0 and critical values as ADF
  - long-run variance $\hat{\lambda}^2 = \hat{\gamma}(0) + 2\sum_{j=1}^{L}\left(1 - \frac{j}{L+1}\right)\hat{\gamma}(j)$, with $\hat{\gamma}(j)$ the autocovariances of the residuals
- **KPSS** (Kwiatkowski–Phillips–Schmidt–Shin) Kwiatkowski et al. (1992): the hypotheses are **reversed**
  - H_0: stationary (around a constant or a trend); H_1: unit root
  - $\mathrm{KPSS} = \sum_{t=1}^{T} S_t^2/(T^2\hat{\lambda}^2)$, S_t the partial sums of the residuals, $\hat{\lambda}^2$ their long-run variance
  - reject stationarity if $\mathrm{KPSS} > 0.463$ (constant) or > 0.146 (trend), at 5%
- Use both: ADF/PP ask "is there evidence against a unit root?"; KPSS asks "is there evidence against stationarity?"
  - ADF does not reject and KPSS rejects: $I(1)$; ADF rejects and KPSS does not: $I(0)$; both or neither reject: inconclusive

The BET: Log Price and Returns



- Log price: ADF does not reject a unit root ($p = 0.736$), KPSS rejects stationarity: $I(1)$
- Returns: ADF rejects, KPSS does not: stationary, $I(0)$; all statistics on the next slide
- The 2008 fall looks like a temporary deviation, but the tests say: permanent shocks, no return to a trend line

SFM_ch7_unit_roots

Unit-Root Tests on Prices and Returns

	log price (constant, trend)			returns (constant)		
	ADF	PP	KPSS	ADF	PP	KPSS
S&P 500	−1.47	−1.57	2.23	−17.61	−105.05	0.12
DAX	−2.69	−2.65	0.94	−23.30	−97.40	0.04
BET	−1.73	−1.54	2.30	−13.65	−74.47	0.12
Bitcoin	−1.48	−1.54	1.81	−20.16	−67.84	0.13
TLV	−3.46	−3.35	0.57	−12.96	−60.79	0.23
SNP	−0.80	−0.74	2.50	−19.50	−62.71	0.29
BRD	−2.18	−2.19	1.64	−59.46	−59.53	0.29
TGN	−0.96	−0.92	0.71	−39.90	−57.83	0.22

- 5% critical values: ADF and PP −3.41 (trend), −2.86 (constant); KPSS 0.146 (trend), 0.463 (constant)
- Log prices: no ADF/PP rejection except TLV (ADF p = 0.044); KPSS rejects stationarity for all: prices are $I(1)$
- Returns: ADF and PP reject strongly; KPSS never rejects (p $>$ 0.10): returns are $I(0)$

SFM_ch7_unit_roots

A Unit Root Is Not Proof of Efficiency

- A unit root in prices is **necessary** for a random walk, but **not sufficient**
  - example: $r_t = 0.3\, r_{t-1} + \varepsilon_t$ makes p_t an $I(1)$ series whose increments are predictable
  - ADF cannot distinguish it from a random walk; the BET has a unit root and autocorrelated returns
- A stationary price (no unit root) would contradict efficiency: prices would return to a predictable level
  - TLV: ADF rejects at 5% with a trend, KPSS also rejects: an inconclusive case, not a profitable rule
- Unit-root tests have low power against ϕ close to 1: the efficiency question is about the increments, and needs tests on returns

The Idea of the Variance Ratio

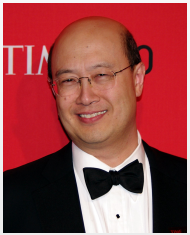
- Under a random walk, $\mathrm{Var}(r_t(q)) = q\,\mathrm{Var}(r_t)$; the **VR** (variance ratio) compares the two sides
  - $\mathrm{VR}(q) = \dfrac{\mathrm{Var}(r_t(q))}{q\,\mathrm{Var}(r_t)} = 1 + 2\sum_{k=1}^{q-1}\left(1 - \dfrac{k}{q}\right)\rho(k)$ Lo & MacKinlay (1988), Campbell, Lo & MacKinlay (1997)
- Reading the number
  - $\mathrm{VR}(q) = 1$: random walk (RW1, RW2, RW3 all give 1)
  - $\mathrm{VR}(q) > 1$: positive autocorrelation, **momentum** (moves continue)
  - $\mathrm{VR}(q) < 1$: negative autocorrelation, **mean reversion** (moves are partly undone)
- One number summarises $q - 1$ autocorrelations with declining weights: more power against persistent, small autocorrelations

Worked Example: VR from the ACF of the S&P 500

- $\hat{\rho}(1), \ldots, \hat{\rho}(4)$ of the S&P 500: -0.079, -0.004, -0.006, -0.022
  - $q = 2$: $\mathrm{VR}(2) = 1 + \hat{\rho}(1) = 0.921$
  - $q = 5$: weights $2(1 - k/5) = 1.6, 1.2, 0.8, 0.4$: $\mathrm{VR}(5) = 1 + 1.6\hat{\rho}(1) + 1.2\hat{\rho}(2) + 0.8\hat{\rho}(3) + 0.4\hat{\rho}(4) = 0.855$
- The Lo–MacKinlay estimator on the same data: $\mathrm{VR}(2) = 0.922$, $\mathrm{VR}(5) = 0.856$
  - almost the same numbers: the two forms differ only by small-sample corrections
- Meaning: the weekly variance is about 0.856 of five daily variances: daily moves of the S&P 500 are partly reversed within a week

Lo and MacKinlay (1988): Estimator and Tests

- With T returns, mean $\hat{\mu}$ and overlapping q-day sums Lo & MacKinlay (1988):
  - $\hat{\sigma}_a^2 = \frac{1}{T-1}\sum_t (r_t - \hat{\mu})^2, \quad \hat{\sigma}_c^2(q) = \frac{1}{m}\sum_t (r_t(q) - q\hat{\mu})^2$
  - $\hat{\sigma}_a^2$: the variance of one-day returns; $\hat{\sigma}_c^2(q)$: the variance of q-day sums, divided by q
  - $m = q(T-q+1)(1-q/T)$ corrects the bias; $\widehat{\mathrm{VR}}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$
- **Homoskedastic** test (RW1): $Z(q) = \dfrac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0,1)$
- **Robust** test (RW3, martingale): $Z^*(q) = \dfrac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{\hat{\theta}(q)/T}} \approx N(0,1)$
  - $\hat{\theta}(q) = \sum_{j=1}^{q-1} [2(q-j)/q]^2 \hat{\delta}(j)$, with the $\hat{\delta}(j)$ of the robust ACF band
  - $\hat{\theta}(q)/T$: the variance of $\widehat{\mathrm{VR}}(q)$ that allows for volatility clustering

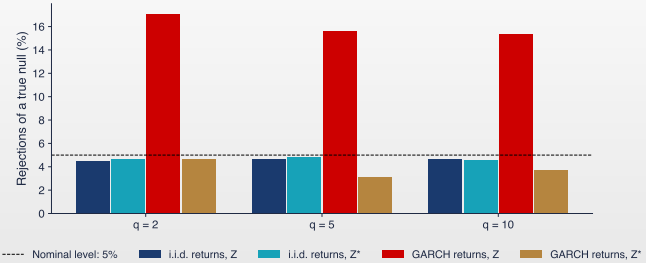


Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0; Wikimedia Commons

SFM_ch7_variance_ratio

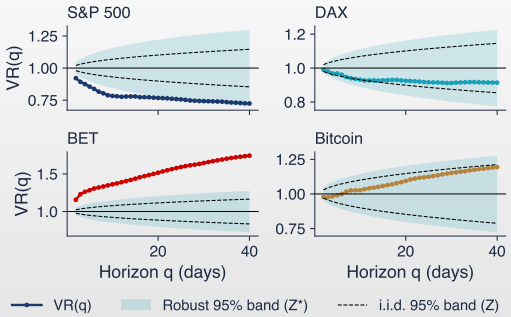
Why the Robust Test: a Monte Carlo Check



- 1 000 samples of $T = 2000$ returns with H_0 true: i.i.d. Normal, and GARCH(1,1) ($\alpha = 0.12$, $\beta = 0.86$)
  - GARCH: uncorrelated returns, with volatility clustering (α: reaction to yesterday's shock, β: persistence)
- i.i.d.: both reject about 5%; GARCH: $Z(2)$ rejects 17.1% of the time, $Z^*(2)$ only 4.7%
- The **size** of a test: how often it rejects a true H_0; Z finds "inefficiency" that is only volatility clustering

SFM_ch7_variance_ratio

Variance Ratios for Horizons of 2 to 40 Days



- S&P 500: VR falls below 1 at every horizon (mean reversion); BET: VR keeps rising with the horizon, 1.517 at 20 days and 1.743 at 40 days (momentum)
- DAX and Bitcoin: inside the robust band (shaded) at every horizon; the i.i.d. band (dashed) is much narrower

SFM_ch7_variance_ratio

Variance Ratios and Robust Tests

	VR(2)	VR(5)	VR(10)	VR(20)	$Z^*(2)$	$Z^*(5)$	$Z^*(10)$	$Z^*(20)$
S&P 500	0.922	0.856	0.783	0.767	−3.37	−2.76	−2.73	−2.06
DAX	0.990	0.968	0.932	0.925	−0.61	−0.88	−1.21	−0.91
BET	1.156	1.283	1.362	1.517	5.99	5.25	4.74	4.94
Bitcoin	0.977	0.991	1.027	1.094	−0.88	−0.19	0.37	0.91
TLV	1.009	0.997	0.954	0.995	0.34	−0.04	−0.48	−0.04
SNP	0.985	0.960	0.975	1.038	−0.53	−0.68	−0.29	0.32
BRD	1.032	1.083	1.078	1.107	1.28	1.55	0.97	0.96
TGN	1.055	1.150	1.142	1.166	2.05	2.60	1.69	1.44

- $|Z^*| > 1.96$: reject the random walk at 5% for that horizon
- S&P 500: significant mean reversion at all four horizons; BET: significant momentum at all four
- Transgaz: $Z^*(2) = 2.05$ and $Z^*(5) = 2.60$; DAX, Bitcoin, TLV, SNP, BRD: no rejection
- But four tests per series, 32 in the table: some "significant" horizons appear by chance

SFM_ch7_variance_ratio

Several Horizons at Once: the Chow–Denning Test

- Testing $q = 2, 5, 10, 20$ separately at 5% gives a much higher chance of at least one false rejection
  - for 4 independent tests: $1 - 0.95^4 = 18.5\%$
- **CD** (Chow–Denning) statistic Chow & Denning (1993): $\mathrm{CD} = \max_i |Z^*(q_i)|$, $i = 1, \ldots, m$
  - critical value from the studentised maximum modulus law: $P(\mathrm{CD} \le c) = (2\Phi(c) - 1)^m$
  - $m = 4$, 5%: $c = 2.49$ instead of 1.96; Φ: the standard Normal distribution function
- Reject the random walk if the largest $|Z^*|$ exceeds 2.49: one decision for all horizons

An Automatic Choice of the Horizon

- The horizon q can be chosen from the data: the automatic VR test Choi (1999)
  - $\mathrm{VR}(k) = 1 + 2\sum_{i=1}^{T-1} w(i/k)\,\hat{\rho}(i)$, with smoothly declining weights w (quadratic spectral kernel) Andrews (1991)
  - the bandwidth k grows with the first-order autocorrelation and with T; statistic $\sqrt{T/k}\,(\mathrm{VR}(k) - 1)/\sqrt{2} \approx N(0,1)$
- **Wild bootstrap** p-value Kim (2009): resample $r_t^* = \eta_t(r_t - \bar{r})$ with $\eta_t \sim N(0,1)$, 500 times
  - each r_t^* keeps the size of r_t (its volatility) but loses any autocorrelation: a robust null distribution
- Three tools, one question: $Z^*(q)$ for a chosen horizon, CD for a set of horizons, the automatic test for a horizon chosen by the data

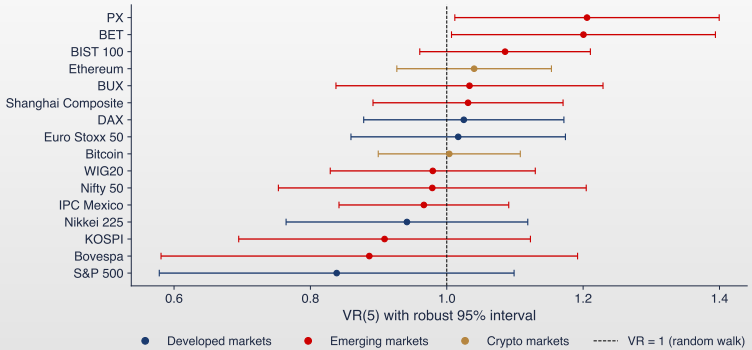
Joint and Automatic Tests on Real Data

	T	CD	p	$\hat{k}$	automatic VR	p (bootstrap)
S&P 500	9 240	3.37	0.003	3.7	−5.21	0.006
DAX	9 288	1.21	0.644	1.7	−0.39	0.608
BET	7 243	5.99	< 0.001	5.6	8.86	< 0.001
Bitcoin	4 384	0.91	0.835	2.0	−1.01	0.352
TLV	3 788	0.48	0.981	1.5	0.31	0.718
SNP	3 815	0.68	0.937	1.7	−0.53	0.562
BRD	3 771	1.55	0.403	2.3	1.58	0.152
TGN	3 725	2.60	0.037	3.0	3.78	0.004

- Both tests reject for the S&P 500, the BET and Transgaz; neither rejects for DAX, Bitcoin, TLV, SNP and BRD
- $\hat{k}$ is large where the first autocorrelation is large (BET 5.6), small where it is near 0 (DAX 1.7)
- Interpretation: rejection does not mean profit
  - an autocorrelation of -0.079 explains $\hat{\rho}(1)^2$, less than 1%, of the variance of tomorrow's return
  - trading costs absorb most of it

SFM_ch7_variance_ratio

Sixteen Markets over the Last Ten Years



- VR(5) with its robust 95% interval, from 19 September 2016 to 18 September 2026: developed, emerging and crypto markets

SFM_ch7_markets_efficiency

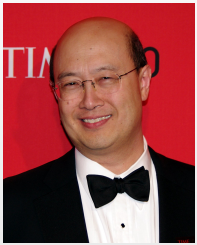
Reading the Cross-Market Comparison

- Only two intervals exclude 1: the PX ($Z^*(5) = 2.08$) and the BET ($Z^*(5) = 2.03$), both above 1
  - two smaller Central European markets, with less liquid index stocks
- **Many markets, many tests**: 16 tests at 5% give 0.8 false rejections on average
  - if the tests were independent, the chance of at least one false rejection would be $1 - 0.95^{16} = 56\%$
  - Chow–Denning over four horizons: no market rejects at 5% (BET p = 0.159, PX p = 0.142)
- S&P 500: VR(5) = 0.84, the lowest, but with a wide interval ($Z^* = -1.22$): the large swings of 2020 inflate $\hat{\theta}$
- Bitcoin and Ethereum: VR(5) close to 1, as for developed markets

SFM_ch7_markets_efficiency

The Adaptive Markets Hypothesis

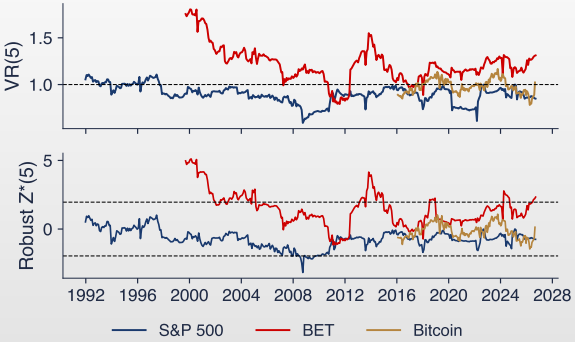
- **AMH** (adaptive markets hypothesis) Lo (2004): markets are ecosystems of traders who learn and adapt
  - profit opportunities appear, are exploited and disappear; efficiency changes with the environment
  - crises, new technology, new participants and regulation move the market away from efficiency or back towards it
- Testable consequence: predictability varies over time
  - evidence: Kim, Shamsuddin & Lim (2011) (U.S., a century of data); Urquhart & McGroarty (2016); survey: Lim & Brooks (2011)
- Tool: the same test on rolling windows



Andrew W. Lo, author of the AMH

Photo: David Shankbone (2012); CC BY 3.0; Wikimedia Commons

Rolling Variance Ratios



⊡ VR(5) and $Z^*(5)$ on windows of 500 trading days (about two years), a new window every 21 days, dated at the window end

SFM_ch7_adaptive_markets

Reading the Rolling Variance Ratios

- BET: VR(5) up to 1.80 in the window ending in 2000; closer to 1 after 2008, apart from short episodes
  - 29% of the windows reject at 5%, all of them with VR > 1 (momentum)
- S&P 500: VR(5) mostly below 1; the lowest, 0.59, in the window ending in 2008
  - 4% of the windows reject, all with VR < 1: reversals are strongest in crises
- Bitcoin: no window rejects since 2016 (185 windows)
- Caution: consecutive windows share 479 of 500 days; the share of rejecting windows is a description, not a test

SFM_ch7_adaptive_markets

Three Sub-Periods per Market

	period	T	$\hat{\rho}(1)$	VR(5)	$Z^*(5)$	CD p
BET	1997–2006	2 306	0.270	1.50	6.15	< 0.001
BET	2007–2016	2 516	0.066	1.08	0.82	0.262
BET	2017–2026	2 421	0.064	1.20	2.04	0.154
S&P 500	1990–2001	3 025	0.017	0.96	−0.62	0.334
S&P 500	2002–2013	3 019	−0.100	0.80	−2.53	0.010
S&P 500	2014–2026	3 196	−0.124	0.85	−1.30	0.080
Bitcoin	2014–2017	1 201	0.028	0.98	−0.23	0.983
Bitcoin	2018–2021	1 461	−0.058	0.98	−0.21	0.374
Bitcoin	2022–2026	1 722	−0.026	1.01	0.18	0.924

- BET: strong momentum in 1997–2006 ($\hat{\rho}(1) = 0.270$), much weaker afterwards
  - since 2017 VR(5) is still above 1, but the joint test does not reject
- S&P 500: the negative autocorrelation appears after 2002
  - Bitcoin: no rejection in any period since 2014; Urquhart (2016) found inefficiency in its earliest years

SFM_ch7_adaptive_markets

Why the Early BET Was Predictable

- **Thin trading**: few trades per day in many index stocks
  - a closing price may be hours old; news enters the index over several days
  - the index then shows positive autocorrelation even if each stock is efficient
- Changes after 2007
  - more liquidity, foreign investors, EU membership, the FTSE Russell upgrade to Secondary Emerging (2020) FTSE Russell (2019)
- Crashes on the Bucharest Stock Exchange: Pele & Mazurencu-Marinescu (2012)



The Stock Exchange Palace, Bucharest

Photo: Neoclassicism Enthusiast (2019); CC BY-SA 4.0; Wikimedia Commons

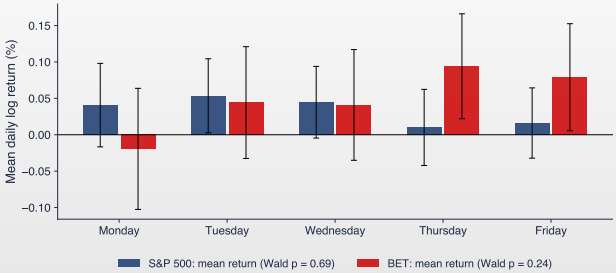
Event Studies and the Strong Form

- **Event study** (semi-strong form) MacKinlay (1997), Campbell, Lo & MacKinlay (1997), Ch. 4: how fast does a price absorb public news?
  - abnormal return: the return minus the return of a model (for example the market); cumulated over a window around the announcement
  - efficient: a jump on the day of the news, no drift afterwards
  - recent research: how fast futures prices absorb news, measured in nanoseconds at Eurex and CME Tak, Schlamp, Osterrieder & Pele (2026); market responses to Ethereum upgrades Lin, Găman, Wang & Pele (2025)
- **Strong form**: can anyone with private information earn excess returns?
  - insiders (managers trading their own shares) do: insider trading is illegal for this reason
  - professional fund managers mostly do not Jensen (1968), Fama & French (2010)

Calendar Anomalies

- An **anomaly**: a pattern in returns that the equilibrium model does not explain
  - **weekend effect**: low or negative Monday returns French (1980)
  - **January effect**: higher returns in January, mostly for small stocks Rozeff & Kinney (1976)
  - **turn-of-the-month effect**: returns concentrated around the first trading days of the month Ariel (1987)
- Do anomalies last?
  - returns of published anomalies fall after publication: traders learn McLean & Pontiff (2016)
  - hundreds of patterns have been tested: a new one needs $|t| > 3$, not 2 Harvey, Liu & Zhu (2016)
- Data mining: with enough calendar splits, some will look significant by chance

Day-of-Week Returns: S&P 500 and BET



- Mean daily return by weekday with 95% intervals (Newey–West standard errors Newey & West (1987))
  - Wald test of equal means across weekdays: S&P 500 p = 0.69, BET p = 0.24
- BET Monday: −0.019%, the only negative mean, yet not significant
  - January: BET 0.185% per day against 0.036% ($t = 1.72$, p = 0.09); S&P 500: p = 0.59
- No calendar effect survives at 5% in our data

SFM_ch7_calendar_anomalies

An Open Question

- **Did the BET become more efficient after Romania's upgrade to Secondary Emerging market (September 2020)?** FTSE Russell (2019)
  - five years before: $\hat{\rho}(1) = -0.002$, $\mathrm{VR}(5) = 1.12$ ($Z^* = 0.80$)
  - five years after: $\hat{\rho}(1) = 0.087$, $\mathrm{VR}(5) = 1.18$ ($Z^* = 1.78$)
  - difference of the two VR(5): $z = 0.34$: no evidence of a change
- Why it is open: COVID-19, new large listings and higher volumes happened at the same time
  - is the index or each stock the right unit?
- AI tools can speed up such a study; they do not replace checking it Wang et al. (2023)

SFM_ch7_adaptive_markets

How AI Could Help

- **Literature**: list studies of efficiency in Central and Eastern European markets and the tests they use
- **Code**: a first draft of rolling Lo–MacKinlay and automatic VR tests for BET stocks, with wild-bootstrap p-values
- **Robustness**: other windows, other horizons, BET-TR instead of BET, trading volume as a control
- Example prompt
  - `Write a Python function that computes the Lo-MacKinlay variance ratio VR(5) and the heteroskedasticity-robust z-statistic for daily BET log returns on rolling 500-day windows, and test whether the mean VR differs before and after 21 September 2020.`

What to Check

- The statistic: robust Z^*, not the homoskedastic Z; a draft that uses Z will "find" inefficiency
- Log returns of the index, not price levels; no unit-root test on returns used as an efficiency test
- Rolling windows overlap: a t-test on the mean of rolling statistics treats dependent numbers as independent
- The event date and the window are fixed before looking at the result
- References: every cited paper must exist; check the DOI

Project Idea

- **Question**: is the Bucharest market more efficient today than in 2015?
  - a second question: does the change appear in the index or in the individual stocks?
  - data: BET since 1997, BET-TR since 2014, BVB stocks (TLV, SNP, BRD, TGN) since 2010 (EODHD)
- Steps
  - rolling VR(2), VR(5) with Z^* and the Chow–Denning test for the index and for each stock
  - compare 2015–2020 with 2020–2025 by a block bootstrap of the difference
  - repeat for the WIG20 and the BUX as controls: did all of them change at the same time?
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

Key Takeaways

- Efficient market: prices reflect information; weak form = past prices cannot be used to earn excess returns
- The right null is the martingale (RW3), not RW1: volatility clustering is not inefficiency
- Log prices are $I(1)$ and returns $I(0)$ in all our series; a unit root alone does not prove efficiency
- Use robust tests: $\tilde{Q}$ and $Z^*(q)$; join horizons with Chow–Denning or let the data choose (automatic VR)
- Evidence: S&P 500 slightly mean-reverting, early BET trending, most markets close to efficient today; efficiency varies over time
- Statistical predictability is not profit: costs and risk decide

Key Formulas

Quantity	Formula
Random walk	$p_t = \mu + p_{t-1} + \varepsilon_t, \quad \mathrm{Var}(r_t(q)) = q\sigma^2$
Sample ACF	$\hat{\rho}(k) = \sum_t e_t e_{t-k} / \sum_t e_t^2, \quad$ band $\pm 1.96\sqrt{\hat{\delta}(k)/T}$
Ljung–Box	$Q_{LB}(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}(k)^2/(T-k) \sim \chi^2(m)$
Runs test	$E[R] = 2n_1 n_2/n + 1, \quad z = (R - E[R])/\mathrm{sd}(R)$
ADF	$\Delta p_t = c + bt + \gamma p_{t-1} + \sum_j \delta_j \Delta p_{t-j} + \varepsilon_t, \quad \tau = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$
KPSS	$\sum_t S_t^2/(T^2\hat{\lambda}^2), \quad H_0$: stationarity
VR	$\mathrm{VR}(q) = 1 + 2\sum_{k=1}^{q-1}(1 - k/q)\rho(k)$
Robust VR test	$Z^*(q) = (\widehat{\mathrm{VR}}(q) - 1)/\sqrt{\hat{\theta}(q)/T}, \quad \hat{\theta}(q) = \sum_{j<q}[2(q-j)/q]^2\hat{\delta}(j)$
Chow–Denning	$\max_i \lvert Z^*(q_i)\rvert, \quad P(\mathrm{CD} \le c) = (2\Phi(c) - 1)^m$

Check Yourself

- **Question**: $\hat{\rho}(1) = 0.10$ and $\hat{\rho}(k) = 0$ for $k \geq 2$; what does VR(2) tell us?
  - **Answer**: $1 + 0.10 = 1.10$: the two-day variance is 10% larger than under a random walk: momentum
- **Question**: ADF does not reject on log prices and KPSS rejects; is the market efficient?
  - **Answer**: we only know that prices are $I(1)$; efficiency needs tests on the increments (ACF, VR)
- **Question**: for the S&P 500 the homoskedastic $Z(q)$ is about twice the robust $Z^*(q)$; which one should we report?
  - **Answer**: $Z^*(q)$: volatility clustering raises the variance of $\widehat{\mathrm{VR}}(q)$, which $Z(q)$ ignores, so $Z(q)$ rejects a true martingale too often
- Next: Chapter 8, volatility estimators and volatility clustering

References (1/3)

- Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- Ariel, R.A. (1987). A monthly effect in stock returns. *Journal of Financial Economics*, 18(1), 161–174.
- Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École normale supérieure*, 17, 21–86.
- Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- Box, G.E.P., Pierce, D.A. (1970). Distribution of residual autocorrelations in autoregressive-integrated moving average time series models. *Journal of the American Statistical Association*, 65(332), 1509–1526.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- Choi, I. (1999). Testing the random walk hypothesis for real exchange rates. *Journal of Applied Econometrics*, 14(3), 293–308.
- Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test. *Journal of Econometrics*, 58(3), 385–401.
- Dickey, D.A., Fuller, W.A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431.
- Escanciano, J.C., Lobato, I.N. (2009). An automatic Portmanteau test for serial correlation. *Journal of Econometrics*, 151(2), 140–149.
- Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.
- Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Fama, E.F. (1991). Efficient capital markets: II. *Journal of Finance*, 46(5), 1575–1617.
- Fama, E.F., French, K.R. (2010). Luck versus skill in the cross-section of mutual fund returns. *Journal of Finance*, 65(5), 1915–1947.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- French, K.R. (1980). Stock returns and the weekend effect. *Journal of Financial Economics*, 8(1), 55–69.

References (2/3)

- FTSE Russell (2019). FTSE Russell announces results of country classification review. Press release, London Stock Exchange Group.
- Granger, C.W.J., Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, 2(2), 111–120.
- Grossman, S.J., Stiglitz, J.E. (1980). On the impossibility of informationally efficient markets. *American Economic Review*, 70(3), 393–408.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Jensen, M.C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2), 389–416.
- Kendall, M.G. (1953). The analysis of economic time-series, Part I: prices. *Journal of the Royal Statistical Society, Series A*, 116(1), 11–34.
- Kim, J.H. (2009). Automatic variance ratio test under conditional heteroskedasticity. *Finance Research Letters*, 6(3), 179–185.
- Kim, J.H., Shamsuddin, A., Lim, K.-P. (2011). Stock return predictability and the adaptive markets hypothesis: evidence from century-long U.S. data. *Journal of Empirical Finance*, 18(5), 868–879.
- Kwiatkowski, D., Phillips, P.C.B., Schmidt, P., Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), 159–178.
- Lim, K.-P., Brooks, R. (2011). The evolution of stock market efficiency over time: a survey of the empirical literature. *Journal of Economic Surveys*, 25(1), 69–108.
- Lin, Y., Găman, A., Wang, Y., Pele, D.T. (2025). Market responses to Ethereum development milestones: an event study approach. SSRN Working Paper 5176758.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lo, A.W. (2004). The adaptive markets hypothesis. *Journal of Portfolio Management*, 30(5), 15–29.
- Lo, A.W. (2017). *Adaptive Markets: Financial Evolution at the Speed of Thought*. Princeton University Press.
- Lo, A.W., MacKinlay, A.C. (1988). Stock market prices do not follow random walks: evidence from a simple specification test. *Review of Financial Studies*, 1(1), 41–66.
- MacKinlay, A.C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13–39.

References (3/3)

- MacKinnon, J.G. (1996). Numerical distribution functions for unit root and cointegration tests. *Journal of Applied Econometrics*, 11(6), 601–618.
- Malkiel, B.G. (2003). The efficient market hypothesis and its critics. *Journal of Economic Perspectives*, 17(1), 59–82.
- McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Pele, D.T., Mazurencu-Marinescu, M. (2012). Modelling stock market crashes: the case of Bucharest Stock Exchange. *Procedia – Social and Behavioral Sciences*, 58, 533–542.
- Phillips, P.C.B., Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75(2), 335–346.
- Roberts, H.V. (1959). Stock-market "patterns" and financial analysis: methodological suggestions. *Journal of Finance*, 14(1), 1–10.
- Rozeff, M.S., Kinney, W.R. (1976). Capital market seasonality: the case of stock returns. *Journal of Financial Economics*, 3(4), 379–402.
- Said, S.E., Dickey, D.A. (1984). Testing for unit roots in autoregressive-moving average models of unknown order. *Biometrika*, 71(3), 599–607.
- Samuelson, P.A. (1965). Proof that properly anticipated prices fluctuate randomly. *Industrial Management Review*, 6(2), 41–49; reprinted in the *World Scientific Handbook in Financial Economics Series*, 25–38.
- Tak, A., Schlamp, S., Osterrieder, J., Pele, D.T. (2026). Racing the news: nanosecond market efficiency at Eurex and CME. SSRN Working Paper 7157232.
- Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80–82.
- Urquhart, A., McGroarty, F. (2016). Are stock markets really efficient? Evidence of the adaptive market hypothesis. *International Review of Financial Analysis*, 47, 39–49.
- Wald, A., Wolfowitz, J. (1940). On a test whether two samples are from the same population. *Annals of Mathematical Statistics*, 11(2), 147–162.
- Wang, H., Fu, T., Du, Y., Gao, W., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620, 47–60.